4.2 Position, Velocity and Acceleration

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How do position, velocity and acceleration come from one another?

Targets

- I can find velocity and acceleration from a position function.
- I can say which way a particle moves from the sign of v.
- I can tell if a particle speeds up or slows down.

Concepts

- **Position, velocity, acceleration** (v = s?, a = v? = s?): Velocity is the derivative of position. Acceleration is the derivative of velocity.
- **Direction** (sign of v = direction): $v > 0$: moving in the positive direction (right). $v < 0$: moving in the negative direction (left). $v = 0$: at rest for an instant.
- **Speed** (speed = $|v|$): Speed is how fast, with no direction. It is the absolute value of velocity.
- **Speeding up or slowing down** ($v \cdot a > 0$ speeding up, $v \cdot a < 0$ slowing down): Same sign for v and a: speeding up. Opposite signs: slowing down.

What the class predicted, and what is true

- A particle has position $s(t)$ meters at t seconds. Its velocity is... **the derivative $s'(t)$.** Velocity is how fast the position changes: the derivative of position.
- A particle moves on a horizontal line, with right as the positive direction. Its velocity is -4 meters per second. It is moving... **left.** Negative velocity means moving in the negative direction: left.

Huddle notes (answers filled in)

Definitions

- **Velocity and acceleration:** v(t) = s?(t). a(t) = **v?(t)**, the derivative of velocity, so a(t) = s?(t).
- **Direction:** If $v > 0$ the particle moves **right**. If $v < 0$ it moves **left**. If $v = 0$ it is **at rest**.
- **Speed:** Speed is the **absolute value** of velocity: speed = $|v|$.
- **Speeding up or slowing down:** v and a with the **same** sign: speeding up. With **opposite** signs: slowing down.

Worked example

A particle has position $s(t) = t^3 - 6t^2 + 9t$. Find v and a, then describe $t = 2$ and $t = 4$.

- Velocity: $v(t) = 3t^2 - 12t + \mathbf{9}$

- Acceleration: a(t) = **6t − 12**
- At t = 2: v = –3 and a = **0**, so it moves left at steady speed for an instant.
- At t = 4: v = 9 and a = 12, both **positive**, so it is speeding up.

Label the picture

1. The dot: the particle at position **s**.
2. The green arrow: the **velocity**.
3. The orange arrow: the **acceleration**.

One worked sample of each skill

Find the velocity

Level 1. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $9t^2 + 8t + 4$ $v(t) = s'(t) = 9t^2 + 8t + 4$.

Level 2. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $8\cos(4t)$ $v(t) = s'(t) = 8\cos(4t)$.

Level 3. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $4t\left(t^2 + 1\right)$ $v(t) = s'(t) = 4t(t^2 + 1)$.

Level 4. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $(1 + 3t)\, e^{3t}$ $v(t) = s'(t) = (1 + 3t)e^{3t}$.

Find the acceleration

Level 1. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $12t$ $v(t) = 6t^2 + 7$. Differentiating v gives the acceleration, the answer above.

Level 2. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $(-24)\cos(2t)$ $v(t) = -12\sin(2t)$. Differentiating v gives the acceleration, the answer above.

Level 3. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $\left(2 + 4t^2\right) e^{t^2}$ $v(t) = 2t\, e^{t^2}$. Differentiating v gives the acceleration, the answer above.

Level 4. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $2\cos(t) - t\sin(t)$ $v(t) = \sin t + t\cos t$. Differentiating v gives the acceleration, the answer above.

Which way is it moving?

Level 1. A particle moves on a horizontal line, with right as the positive direction. At $t = 3$ seconds its velocity is -2 meters per second. Which way is it moving?
(A) to the left (B) to the right (C) cannot tell (D) at rest for an instant
Answer: A The velocity is -2, so the particle is to the left.

Level 2. A particle moves on a horizontal line, with right as the positive direction. Its velocity is $v(t) = 3t^2 + 4t - 1$ meters per second. At $t = 2$ which way is it moving?
(A) to the left (B) at rest for an instant (C) cannot tell **(D) to the right**
Answer: D $v(2) = 19$, so the particle is to the right.

Level 3. A particle moves on a horizontal line, with right as the positive direction. Its position is $s(t) = t^3 - 4t^2 - 4t + 5$ meters. At $t = 4$ which way is it moving?
(A) at rest for an instant (B) cannot tell (C) to the left **(D) to the right**
Answer: D $v(t) = 3t^2 - 8t - 4$ and $v(4) = 12$, so the particle is to the right.

Level 4. A particle moves on a horizontal line, with right as the positive direction. Its position is $s(t) = 3t^3 - 8t^2 + 11t + 5$ meters. At $t = 4$ which way is it moving?
(A) cannot tell (B) to the left (C) at rest for an instant **(D) to the right**
Answer: D $v(t) = 9t^2 - 16t + 11$ and $v(4) = 91$, so the particle is to the right.

Speeding up or slowing down?

Level 1. At one moment a particle has velocity $v = -3$ and acceleration $a = -4$ (meters per second, and meters per second squared). Is it speeding up or slowing down?
(A) speeding up (B) slowing down (C) neither: its speed is not changing at that moment
Answer: A $v = -3$ and $a = -4$ have the same sign, so it is speeding up.

Level 2. A particle has velocity $v(t) = 3t^2 + 2t - 1$ meters per second. At $t = 0$ is it speeding up or slowing down?
(A) neither: its speed is not changing at that moment **(B) slowing down** (C) speeding up
Answer: B $v(0) = -1$ and $a(0) = 2$. Opposite signs: slowing down.

Level 3. A particle has position $s(t) = t^3 - 4t^2 - 4t + 5$ meters. At $t = 5$ is it speeding up or slowing down?
(A) slowing down **(B) speeding up** (C) neither: its speed is not changing at that moment
Answer: B $v(5) = 31$ and $a(5) = 22$. Same sign: speeding up.

Level 4. A particle has position $s(t) = 3t^3 - 8t^2 + 11t + 5$ meters. At $t = 4$ is it speeding up or slowing down?
(A) slowing down **(B) speeding up** (C) neither: its speed is not changing at that moment
Answer: B $v(4) = 91$ and $a(4) = 56$. Same sign: speeding up.